

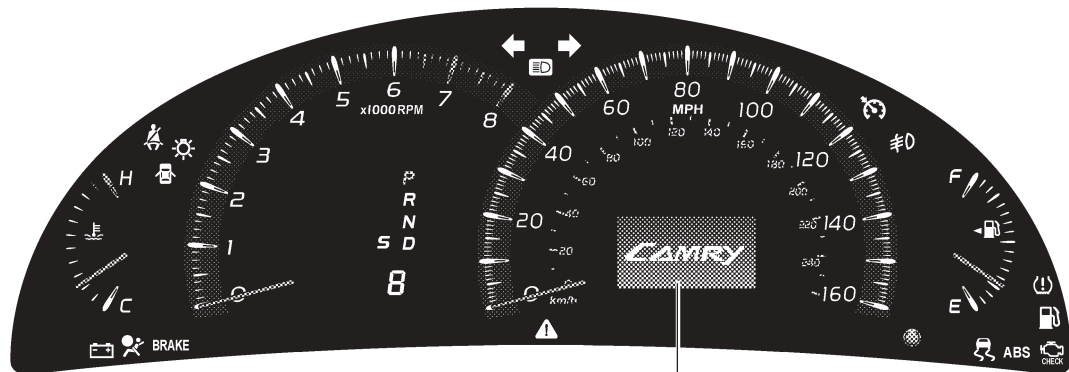
METER

■ COMBINATION METER

1. General

- An optitron display type combination meter is used on XLE and LE grade models. The optitron display type meter realizes excellent visibility through the use of smoke acrylic in the protective panel, and bright LEDs (Light Emitting Diodes) that have high contrast to illuminate the indicator and the dial.
- A multi-information display is provided on the speedometer on XLE grade models.
- A meter ECU and buzzer are enclosed in the combination meter. This ECU maintains communication with other ECUs through the CAN (Controller Area Network).
- Illumination control, which turns on the combination meter illumination at different time intervals when the power source* is switched to IG-ON, has been provided.
- A step-motor type movement is used to actuate the indicators of the speedometer, the fuel gauge, the engine coolant temperature gauge and the tachometer.

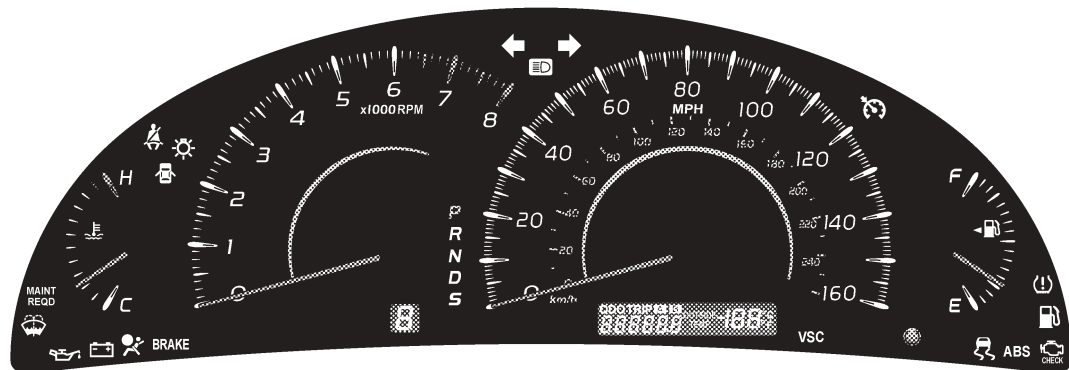
*: The power source condition can be changed by operating the engine switch on models with the smart key system, and the ignition switch on models without the smart key system.



Multi-information Display

025BE13Y

XLE Grade Models with 2GR-FE Engine for U.S.A.



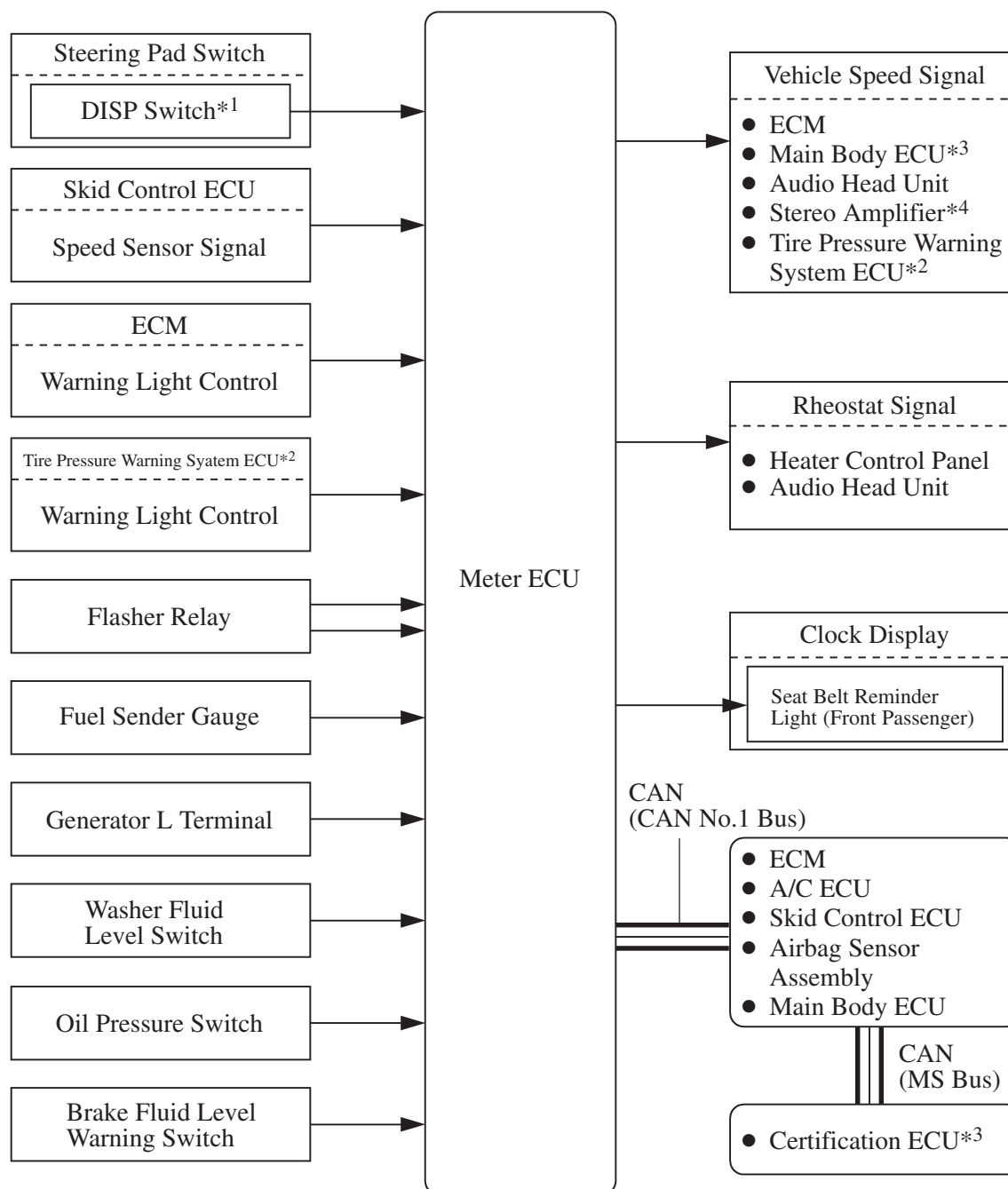
LE Grade Models with 2GR-FE Engine for U.S.A.

025BE52Y

Service Tip

If the LEDs malfunction, the entire combination meter assembly must be replaced. Refer to the 2007 Camry Repair Manual (Pub. No. RM0250U).

2. System Diagram



025BE14P

*1: Only for XLE grade models

*2: Only for U.S.A. models

*3: Only for models with smart key system

*4: Only for models with JBL premium sound system

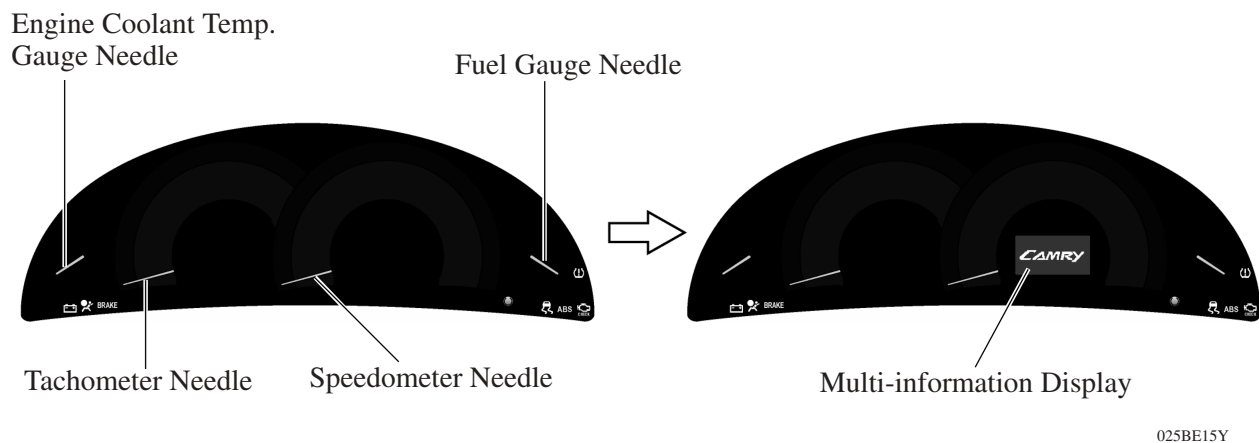
► Input and output communication signals of the combination meter ◀

Protocol	ECU	Input Signal	Output Signal
CAN (CAN No.1 Bus)	ECM	● Engine speed ● Engine coolant temperature ● Fuel injection volume ● Starter condition ● Shift position ● S mode indicator ● Current range position ● Buzzer sounding request ● Indicator light control ● Diagnosis (Cruise) ● Engine type information	—
	A/C ECU	Outside temperature	Vehicle speed
	Airbag Sensor Assembly	● Warning light control ● Seat belt remainder control (D, P) ● Diagnosis	Vehicle speed
	Skid Control ECU	● Warning light control ● Indicator light control ● Diagnosis	—
	Main Body ECU	● Lighting status ● Parking brake switch ● Courtesy switch ● Buzzer sounding request ● Key unlock warning switch ● Warning display control ● Diagnosis	—
CAN (MS Bus)	Certification ECU*	● Warning display control ● Buzzer sounding request	—

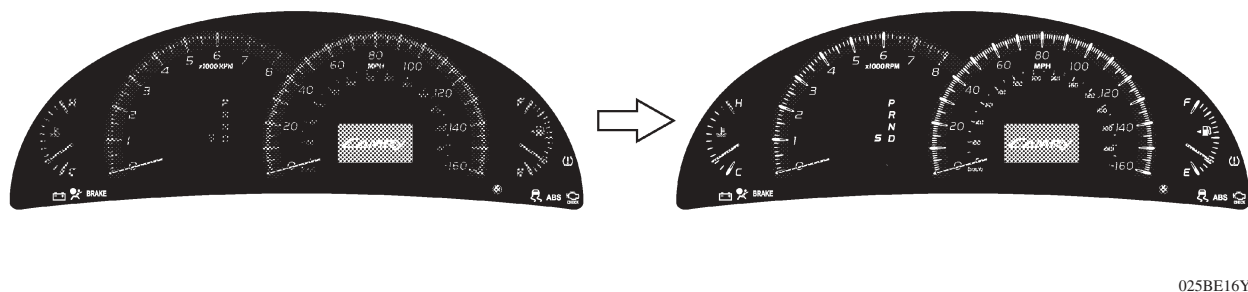
*: Only for models with smart key system

3. Illumination Control

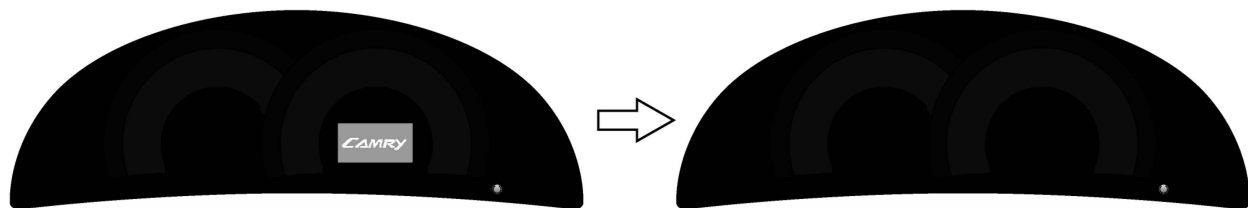
- When the power source is switched to IG-ON, the illumination control operates as follows (1, 2). However, all illuminations turn off while the engine is cranked.
 - When the power source is switched to OFF, the illumination control is as follows (3).
- 1) The needles of the speedometer, tachometer, fuel gauge and engine coolant temperature gauge are illuminated and “Camry” appears on the multi-information display.



- 2) After the multi-information display illuminates, the meter illumination gradually fades in.



- 3) All illuminations other than the multi-information display go off, and “Camry” appears on the multi-information display. Then the “Camry” disappears and the multi-information display turns off.



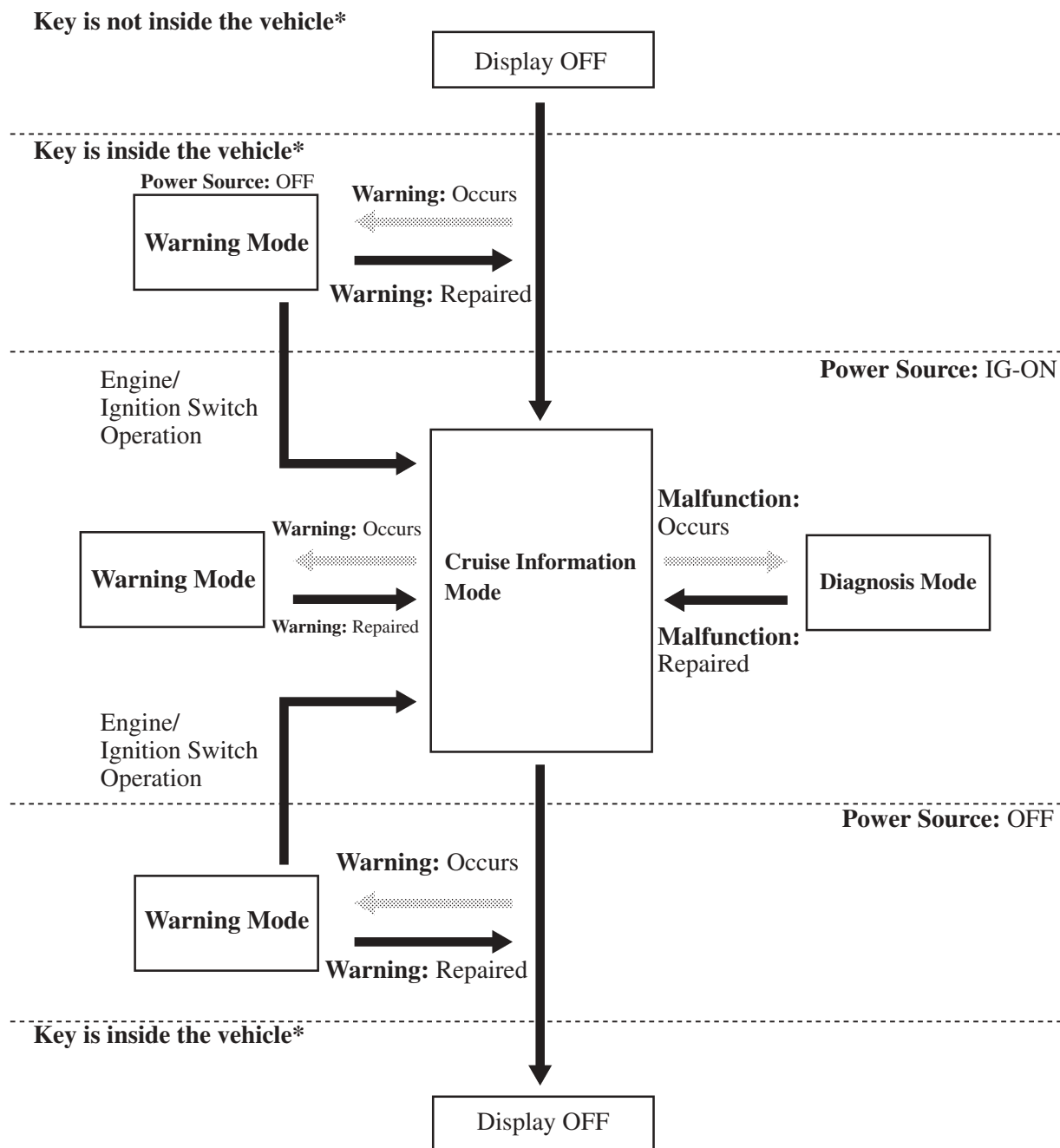
4. Multi-information Display

General

- The multi-information display has three modes:

Mode	Outline
Cruise Information (See page BE-31)	• Five types of information can be displayed: outside temperature, driving range, average fuel consumption since refueling, distance driven since engine start, and average speed since engine start. • The display can be switched by using the DISP switch.
Warning (See page BE-32)	Interrupts the multi-information display immediately when a warning occurs.
Diagnosis (See page CH-137)	DTC (Diagnostic Trouble Code) for the brake control system (TRAC and VSC) can be displayed.

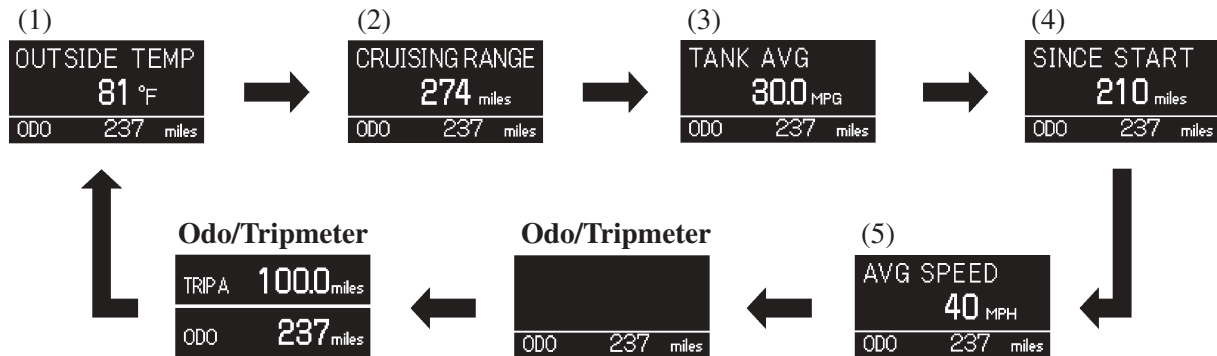
- The multi-information display modes can be switched as shown in the flow chart below:



*: Models with smart key system

Cruise Information Mode

The cruise information is displayed in the following order, changing each time the DISP switch is pressed. However, pressing the DISP switch for approximately 1 second or more changes the display to the outside temperature indication.



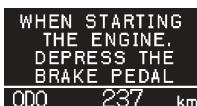
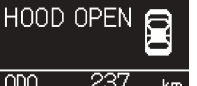
025BE18P

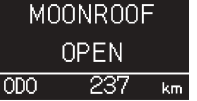
Information		Outline
(1)	Outside Temperature	Displays the outside temperature in accordance with the outside temperature sensor signal from the A/C ECU.
(2)	Driving Range	- Displays the range, calculated by the combination meter which continuously monitors and stores fuel consumption data and the residual fuel volume when IG-ON has been selected. - Updated every 1 mile (U.S.A. model) or 1 km (except U.S.A. model [package option]).
(3)	Average Fuel Consumption Since Refueling	- Displays the value, calculated by the combination meter, based on the distance driven since refueling and the fuel consumption volume (fuel injection signals from the No.1 injector). - The combination meter determines the vehicle has been refueled through the signal from the fuel sender gauge. - Updated every 10 seconds.
(4)	Distance Driven Since Engine Start	- Displays the value calculated by the combination meter based on the distance driven since engine start. - Updated every 1 mile (U.S.A. model) or 1 km (except U.S.A. model [package option]).
(5)	Average Speed Since Engine Start	- Displays the average speed calculated by the combination meter based on the length of time and the distance driven since engine start. - Updated every 10 seconds.

Warning Mode

1) General

- When a warning is necessary, the warning display interrupts the multi-information display.
- The master warning light may illuminate or flash and the buzzer may sound depending on the item in the multi-information display.

Warning	Detail	Warning	Detail
  : Flash Buzzer: Sound 025BE19P	The key is not inside the vehicle. (Models with smart key system)	  : Flash Buzzer: Sound 025BE20P	The driver door is opened with the shift lever in any position other than P. (Models with smart key system)
  : Flash 025BE21P	Steering lock has not been released. (Models with smart key system)	  : Flash 025BE22P	Steering lock is malfunctioning. (Models with smart key system)
 Buzzer: Sound 025BE23P	The power source is switched from OFF to ACC twice with the brake pedal released. (Models with smart key system)	  : Illuminate Buzzer: Sound 025BE24P	TRAC and VSC are malfunctioning. (Models with Brake Control System [ABS with EBD, Brake Assist, TRAC and VSC])
  : Flash Buzzer: Sound 025BE25P	Key battery is low. (Models with smart key system)	  : Flash Buzzer: Sound 025BE26P	Parking brake is still engaged with the vehicle having reached a speed of 5 km/h (3 mph).
     : Flash*  : Illuminate Buzzer: Sound* 025BE27P	Any door is open. *: The vehicle having reached a speed of 5 km/h (3 mph).	  : Flash* Buzzer: Sound* 025BE28P	Engine hood is open. *: The vehicle having reached a speed of 5 km/h (3 mph).
		  : Flash*  : Illuminate Buzzer: Sound* 025BE29P	Luggage compartment door is open. *: The vehicle having reached a speed of 5 km/h (3 mph).

Warning	Detail	Warning	Detail
 ! : Flash Buzzer: Sound 025BE30P	Engine oil pressure is low.	 ! : Flash Buzzer: Sound 025BE31P	Engine coolant temperature is high.
 ! : Illuminate Buzzer: Sound 025BE32P	Washer fluid level is low.	 Displayed for 8 sec. ! : Flash Buzzer: Sound 025BE33P	The sliding roof is open and the driver door is open.
 ! : Illuminate Buzzer: Sound 025BE34P	Comes on approximately 4500 miles after the engine oil is changed. (Only for U.S.A. models)	 ! : Illuminate Buzzer: Sound 025BE35P	Comes on approximately 5000 miles after the engine oil has been changed. (Only for U.S.A. models)

2) Oil Replacement Reminder

- The oil replacement reminder appears to remind the driver to change the engine oil in accordance with the vehicle driving distance. This reminder is not provided on models equipped with destination package options (reminder provided only on U.S.A. models).
- The meter ECU calculates the vehicle driving distance based on the signals from the skid control ECU.
- There are two types of warnings: one is displayed when the vehicle driving distance has reached 4,500 miles or more since the last time the system was reset, and the other is displayed when the driving distance has reached 5,000 miles or more.
- The “OIL MAINT REQD SOON” warning appears for approximately 15 seconds after the power source is changed to IG-ON, and then goes off.
- The “OIL MAINT REQD” warning remains on while the power source is IG-ON.



4,500 miles or more



5,000 miles or more

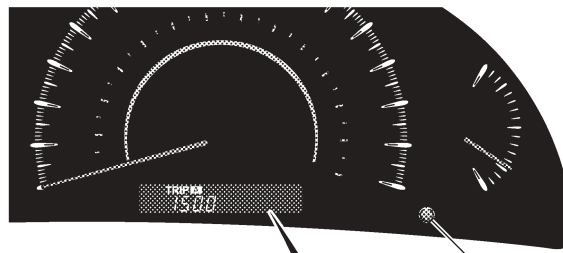
025BE36P

- After the engine oil has been changed, the accumulated vehicle driving distance is memorized in the meter ECU and should be reset through the operation of the “TRIP” meter reset knob. At this point, the accumulated vehicle driving distance is reset to zero.

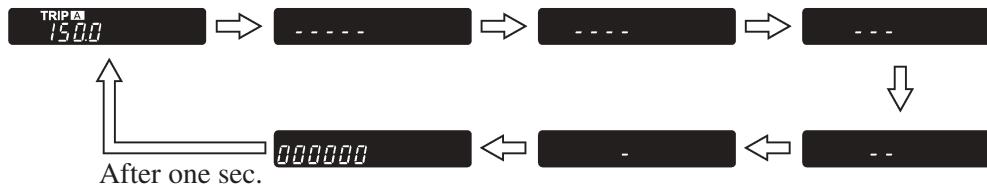
Service Tip

The accumulated vehicle driving distance is stored in the meter ECU and can be reset using the following procedure.

- 1) Switch the power source to IG-ON and make sure that the LCD of the tripmeter/TRIP A (SE and LE grade) or LCD of the multi-information display/TRIP A display (XLE grade) is on.
- 2) Switch the power source to OFF. While pushing the “TRIP” meter reset knob, switch the power source to IG-ON.
- 3) With the power source in the IG-ON mode, keep holding the “TRIP” meter reset knob (for at least five seconds) with the LCD counting down as shown below. Release the “TRIP” meter reset knob when the resetting is complete.
- 4) When the resetting is complete, the LCD displays “000000” (SE and LE grades), or the LCD displays “COMPLETE” for 1 second, the master warning light illuminates and the buzzer sounds once (XLE grade). Then, the LCD displays the odometer.

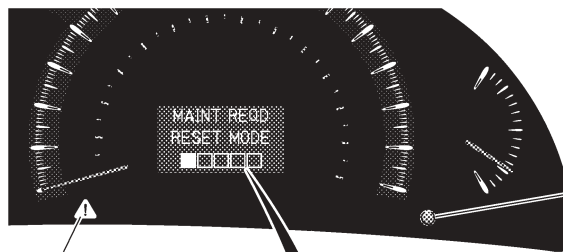


TRIP Meter Reset Knob



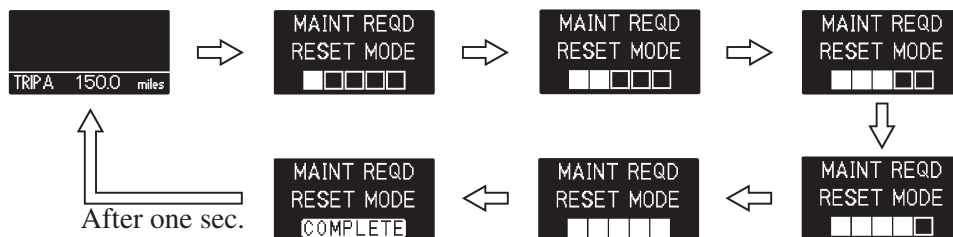
LE Grade Models

025BE37Y



Master Warning Light

TRIP Meter Reset Knob



XLE Grade Models

025BE38Y

5. Buzzer

General

The table below shows the warning and reminder functions of the buzzer.

Function	Item
Warning	● Multi-information Display Warning Mode Indication (See page BE-32) ● Shift Down Warning (See page CH-73)
Reminder	● Key Reminder (For models with smart key system, see page BE-119.) ● Seat Belt Reminder (See page BE-158)

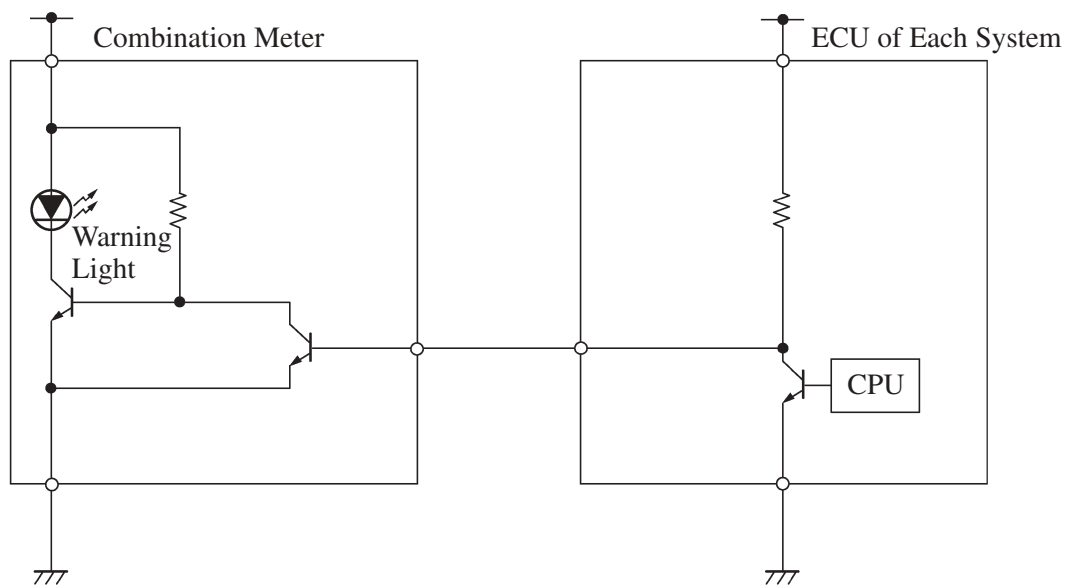
Key Reminder (Models without Smart Key System)

When the driver's door is opened with the ignition key in the ACC or LOCK position, the meter ECU sounds the buzzer to warn the driver that the ignition key has not been removed.

6. Active Circuit

Active circuits are used in the tire pressure warning light circuit in order to illuminate the light when there is an open or short circuit in the wiring harness. Thus, the malfunction detection area has been expanded.

► Active Circuit Conceptual Drawing ◀



264BE56